TSE | THAMMASAT SCHOOL OF ENGINEERING

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VisionQC

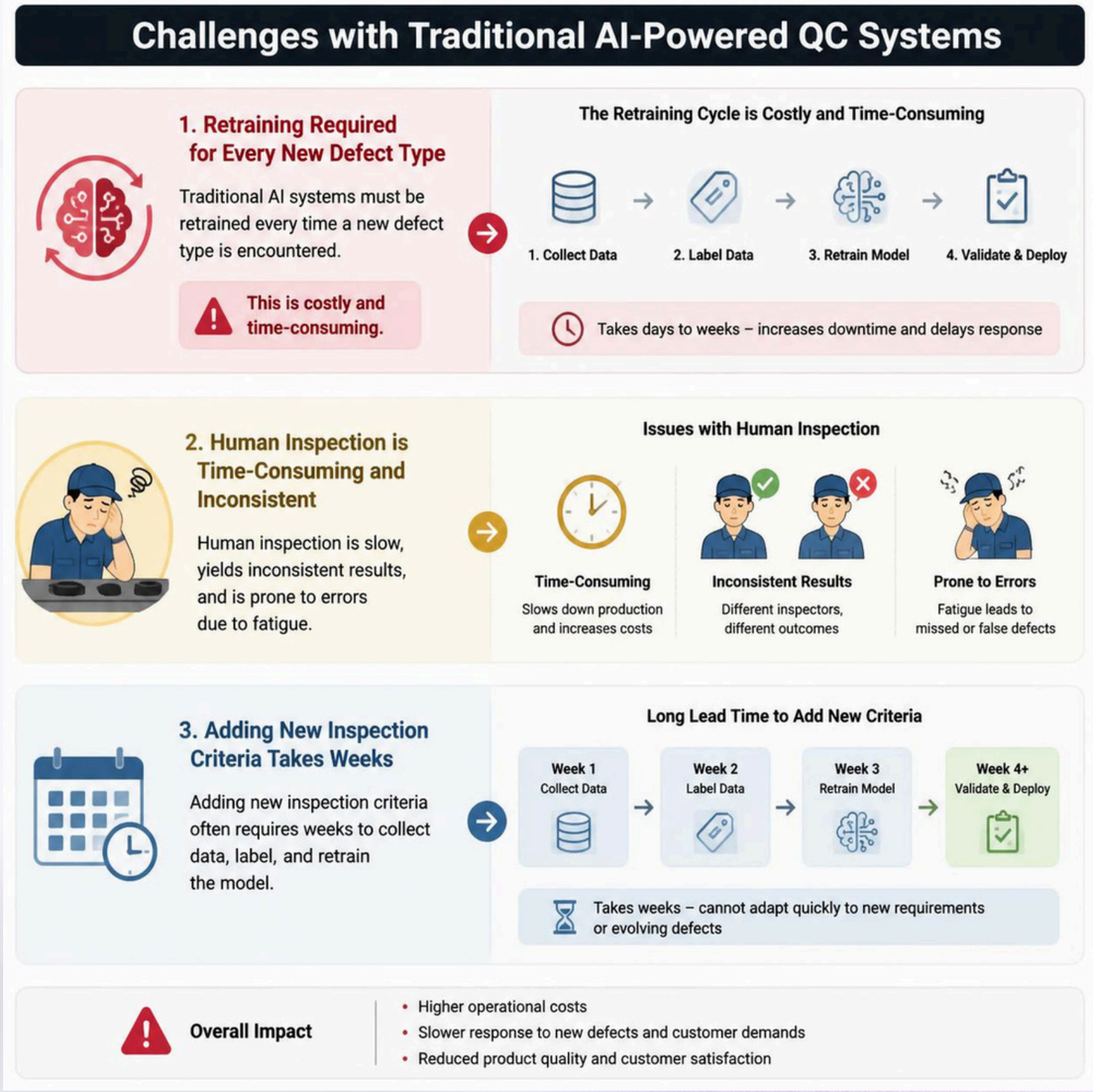
AN AI-POWERED INSPECTION SYSTEM FOR ACCURATE AND EFFICIENT DEFECT DETECTION

Project Overview

An AI-powered quality inspection system that adapts to new defect types instantly just by changing a text prompt, no retraining required. Vision AI, workflow automation, and embedded hardware work together to detect, display, and reject defective products in real time, all monitored through an immersive HUD interface.

PainPoint

- Traditional AI-based QC systems require retraining every time a new defect type appears — costly and time-consuming
- Manual inspection is slow, inconsistent, and prone to human fatigue/error
- Adding a new inspection criterion often means weeks of data collection, labeling, and model retraining

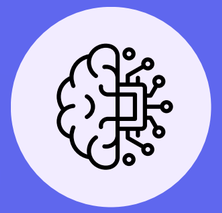


Solution

- Open-vocabulary Vision AI detects new defect types instantly via text prompts — no retraining needed
- Immersive HUD lets operators update inspection targets on the spot, directly from the production floor

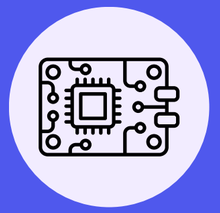
System Architecture

Hardware



Raspberry Pi

- Streams a live video feed directly to the Pico 4 Web HUD.
- Captures and sends high-resolution photos to the n8n server.



ESP32

- Serves as the main hardware interface.
- Receives data from connected IR or Photoelectric sensors
- Controls actuators like servo motors, solenoids, or relays to reject defective items, and triggers warning LEDs or buzzers.

Pico 4 VR Headset

- Acts as the primary immersive display device for the operator.

System Architecture

Software

n8n

- Central middleware between Raspberry Pi and web interface
- Evaluates AI responses
- Log system statistics

NVIDIA LocateAnything

- Processes images based on prompt texts
- Return Bounding box coordinates to the n8n server

Edge Device Firmware

- Python webserver on Raspberry Pi
- C++ via Arduino IDE on ESP32

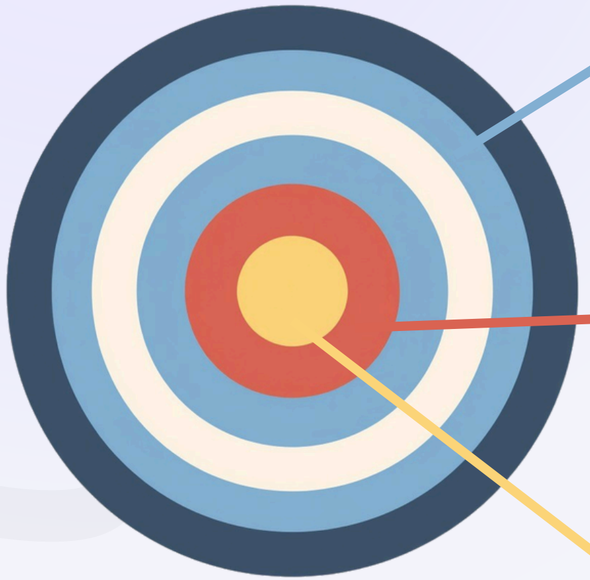
PICO 4 Web Frontend

- Built with HTML5, CSS, and JS
- Includes a Prompt box to input defect targets

Supabase

- Stores inspection results and system logs
- Saves pass/fail status for each detected defect
- Provides historical records for analysis and reporting

Key Features



1

Prompt-based defect detection

Instantly change inspection targets via text, no AI retraining needed

2

Live data logging

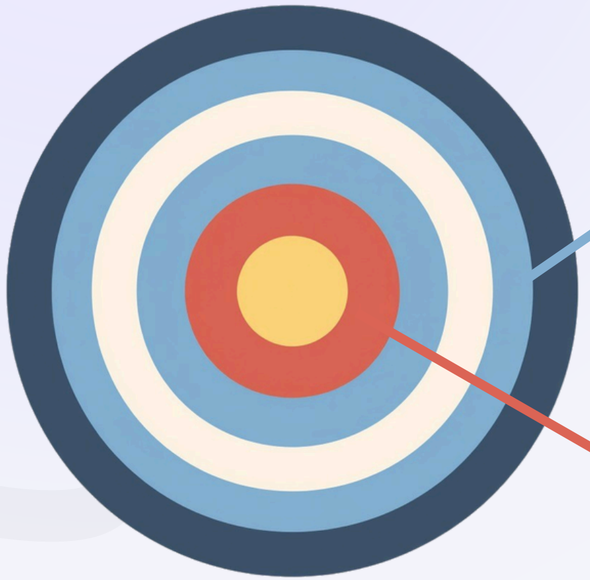
Every detection is recorded with timestamp, defect type, and confidence score for analysis

3

Automated workflow

n8n orchestrates the entire pipeline, making it easy to modify or extend without rewriting code

Key Features



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Embedded hardware

Enables fast, low-cost physical automation for defect rejection

5

Immersive HUD

Displays live video with AI-detected bounding boxes and lets operators update prompts instantly

System Architecture

System flow

01

Camera + Raspberry Pi

Streams live video and captures a frame on trigger or timed interval

02

n8n

receives images, attaches the current text prompt, sends requests to LocateAnything

03

NVIDIA LocateAnything

Performs open-vocabulary object detection on the received image and prompt

04

Pico 4 (Web HUD)

Displays the live video feed with AI-detected bounding boxes overlaid in real time

05

ESP32 (Embedded Controller)

controls the servo/relay to physically remove defective items

06

Database (supabase)

Logs inspection results timestamp, defect type, and confidence score for tracking and reporting